Proposal briefing: an offline medical assistant for places with little internet

Prepared 19 October 2026 · For a project launch proposal · Planning document, not a clinical protocol or a legal opinion

This briefing answers four questions for the launch proposal: how leading hospitals use artificial intelligence today; whether a person can diagnose a condition and choose a medicine from blood tests without a physician; how many people one application can serve; and what training is required to answer in another language.

> **Recommendation.** Launch an offline aid for a clinic, pharmacy, or health program. The patient describes symptoms and can attach a lab sheet. The application explains the information, shows the local guideline it used, and lists questions for the visit. A licensed nurse, pharmacist, or doctor decides whether any medicine is appropriate. That is the pattern already used in major hospitals, and it is the product that can be defended.

1. How leading hospitals use artificial intelligence

Top hospitals use AI to speed work that a clinician still signs. They do not hand a patient a model and let it choose a medicine. Four current examples:

- **Mayo Clinic** uses generative AI to read referral documents, summarize them, and flag urgent cases so staff can check the summary and contact high-risk patients sooner. In the laboratory, AI reads labels and paperwork and proposes where a specimen should go; a technologist confirms the fields before the sample moves. Mayo has also put 17 radiology algorithms into clinical use and watches them for failures and data drift. In its telestroke network, imaging software such as Brainomix marks suspected stroke findings for the stroke physician. The physician still makes the treatment decision.
- **Cleveland Clinic** deployed an ambient scribe to more than 4,000 ambulatory clinicians in four months. By a year later, more than 4,800 clinicians had used it across more than 3.5 million encounters. The tool drafts the visit note. The clinician reviews, edits, and signs it.
- **Mass General Brigham** reports ambient documentation in use by about 4,000 clinicians, with about a 20 percent reduction in administrative time. Imaging tools pass an AI governance committee, are tested on local patients, and stay under monitoring after go-live.
- **Kaiser Permanente** requires each tool to show a benefit in safety, accuracy, and equity on its own member data, then keeps measuring outcomes after launch. Its Advance Alert Monitor, used in Northern California hospitals, warns staff about patients who may deteriorate. Staff act on the warning.

The shared design is the same in each case. The model drafts, sorts, or highlights. A named professional accepts or overrides the result. Hospitals also log errors and retest when the pictures or the paperwork change.

2. Can a person diagnose themselves and choose a medicine?

A model can read a blood-test sheet and produce sentences that look like a diagnosis and a drug list. That output is not a diagnosis, and it is not a prescription. [BioMistral](https://huggingface.co/BioMistral/BioMistral-7B), MedGemma, and Med42 are published as research and developer models. BioMistral's own model card says it has not been tested in real clinics and should not be used for professional health decisions. MedGemma's card says its output is not for a clinical diagnosis or a treatment decision.

> **Proposal position.** Do not claim that a person can diagnose their own condition and select a medication from laboratory results without a physician. Leading hospitals do not offer that service. In the United States, software intended to diagnose or to recommend a specific treatment is generally a medical device, and the January 2026 FDA clinical-decision-support guidance keeps the clinician able to review the basis of any recommendation. Other countries have their own device and pharmacy rules. A local regulatory adviser has to classify the product before it is offered to the public.

What the same model can usefully do, and what the proposal should promise:

- Explain a blood test in everyday language: which number is high or low, and what that number usually refers to.
- List possibilities and the warning signs that mean emergency care, and show the guideline sentence each point came from.
- Prepare questions for the clinician and a food or self-care note that a local clinician has already approved.
- For a logged-in nurse or pharmacist, show medicines that are already on the local essential-medicines list, with the source, and require that person to accept or reject each suggestion.

The benefit of the model is faster, more consistent preparation in a place with no network. The benefit is not a substitute for the person who is allowed to diagnose and prescribe.

3. How many users one application can serve

The limit is the computer that runs the model, not the number of names in a database. An ordinary application database can hold tens of thousands of accounts. Only a few of those people can receive an answer at the same moment.

Setup	At the same time	In a working day
One clinic laptop running BioMistral 7B or MedGemma 4B	1 active conversation	1 clinician, about 20 to 40 assisted visits if each AI step takes under a minute
One server with a 24 GB graphics card, 7B or 8B model, short questions	About 4 to 8 comfortable chats. More than that and replies slow down	A few hundred short sessions if they are spread through the day
A 70-billion-parameter model on one large card	1 or 2 chats	Leave this size out of the first launch

These are planning ranges, not a measured benchmark of a finished app. A clinic that is often offline should plan on **one computer per clinician**. Each machine holds its own copy of the model. Adding a second laptop adds a second simultaneous user. A shared server helps only at sites that have a local network, even when that network does not reach the internet.

For the proposal, a practical first target is **10 to 20 sites, one workstation each**. That is about 10 to 20 concurrent users and on the order of 200 to 800 assisted encounters a day, depending on how busy the clinics are.

4. Training required for another language

BioMistral was built on English medical papers. Downloading it does not give you a library you can translate. The model file is the result of training, not the papers themselves. Full translation of PubMed Central is a national research project. A clinic launch does not need that.

Stage	What is trained or prepared	People and equipment	Time before a pilot
1. Local documents	National treatment guidelines, the essential-medicines list, and approved food and self-care notes, written in the local language and checked line by line	One bilingual clinician plus a translator. No model retraining	4 to 8 weeks for one condition set
2. Search on the device	The app finds the relevant local paragraph and asks the model to answer from that paragraph only	One developer. The documents stay on the clinic computer	2 to 4 weeks after the documents exist
3. Language fine-tune	About 2,000 to 10,000 checked pairs: a question in the local language and the approved answer	Native-speaking clinicians to write and score the pairs. One computer with 24 GB of graphics memory for a small adapter (LoRA)	About 6 to 12 weeks, including review
4. New foundation model	Continued training on a large translated medical library	A GPU cluster and a cleaned corpus. This is university or ministry work	Not part of the launch

Stage 1 and stage 2 are enough to open a pilot. Stage 3 is justified when the model still answers in English, or mixes languages, after the local documents are in place. A bilingual clinician has to approve every drug name and dose that the app is allowed to show. Machine translation alone is not an acceptable source for those lines.

5. Suggested launch scope

1. One country, one language, and one user type: a nurse or a pharmacist.
2. One set of problems already covered by a national guideline, such as common outpatient illness or childhood illness.
3. BioMistral 7B or MedGemma 4B on a computer with 16 GB of memory. The model file is free. The computer is the cost, roughly $800 to $1,400 for a usable laptop, as priced in October 2026.
4. Red-flag symptoms end the session and name where to refer. The app does not invent a dose.
5. A 10-to-20-site pilot that records how often the clinician agrees, overrides, or rejects a suggestion.

Charge the facility, not the patient: a setup fee for translation, local medicines, and training, then a yearly fee for support and guideline updates delivered by a brief connection or by USB.

6. Sources

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- Mayo Clinic telestroke evaluation of Brainomix imaging support. https://pmc.ncbi.nlm.nih.gov/articles/PMC12163383/
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- U.S. FDA, Clinical Decision Support Software guidance, January 2026. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software
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Figures for concurrent users and visit volume are engineering estimates for planning. They should be replaced with measurements from the pilot. Regulatory class depends on the country of launch and on the exact claims made to patients.